An Automatic Algorithm for Estimating Tropical Cyclone Centers in Synthetic Aperture Radar Imagery

Yan Wang, Gang Zheng, *Senior Member, IEEE*, Xiaofeng Li, *Fellow, IEEE*, Lizhang Zhou, *Member, IEEE*, Bin Liu, *Member, IEEE*, Peng Chen, Lin Ren, and Xiaohui Li

***Abstract*— Synthetic aperture radar (SAR) can monitor the sea surface imprints of tropical cyclones (TCs) with high spatial resolution, day and night. Automatically locating TC center positions in SAR images is a challenging task. This article developed a two-stage, fully automatic TC-center estimation algorithm. First, the sea surface wind directions (SSWDs) at SSWD points are retrieved by the improved local gradient (ILG) method. We incrementally deflected the SSWD outward at a 0.5° angle from −50° to 10° (the negative angles represent clockwise deflection). The heat maps are generated for each of the 121 angles, and the values at each heat map are the cumulative numbers of the lines perpendicular to the compensated SSWDs. The site corresponding to the maximum cumulative number in all 121 heat maps is the coarsely estimated center position. This center search is the culmination if it falls outside the SAR image. Otherwise, the second stage is triggered, and the sub-SAR image (150 km × 150 km) centered at the coarsely estimated center position is extracted. Then, the first-stage procedure is repeated with the sub-SAR image to precisely estimate the center position. Optionally, the precisely estimated center position can be further adjusted by considering that normalized radar cross section (NRCS) is normally minimal at the TC center. We applied the algorithm to 87 SAR images. Five of these images do not contain TC centers. The results are in good agreement with the visually located TC center positions and those in the best track (BT) datasets.**

***Index Terms*— Sea surface wind direction (SSWD), synthetic aperture radar (SAR), tropical cyclone (TC).**

I. Introduction

A TROPICAL cyclone (TC) is a destructive storm system rotating around its low-pressure center [1]. TCs cause strong winds, heavy rains, and storm surges and pose severe threats to human life and property security [2]–[5] when they make landfall. As one of the most critical parameters, the center position is vital in monitoring and forecasting TCs [6]–[13].

Visible and infrared sensors have been widely applied to monitor and study TCs because of their extensive coverage and high imaging rate [12]. Many previous research efforts have been made to locate TC center positions objectively from visible and infrared images. TC centers can be identified from the cues provided by the spiral structure feature in TC images. Consequently, a TC center is determined by fitting a logarithmic spiral curve with the TC spiral cloud patterns from satellite visible and infrared images [14]. The model parameters, including TC center positions, can be optimized by a genetic algorithm [15] and the chaos immune evolutionary algorithm [16]. Other algorithms, such as the ellipse curve model [17], brightness-temperature gradient-based algorithm [18], and weight slice method [19], were also developed based on the morphological features of TCs in infrared images.

Visible and infrared sensors capture the cloud top signatures of TCs. However, a TC system usually has a vertically tilted structure, and consequently, the TC center position near the sea surface and that at the cloud top are uncoordinated [20], [21]. Unlike visible and infrared cameras, in principle, synthetic aperture radar (SAR) can capture the sea surface signatures of TCs due to its cloud-penetrating capability [22]–[25], and the SAR image can be utilized to locate TC center positions at the sea surface. This application of SAR has recently drawn considerable attention, and various objective TC center

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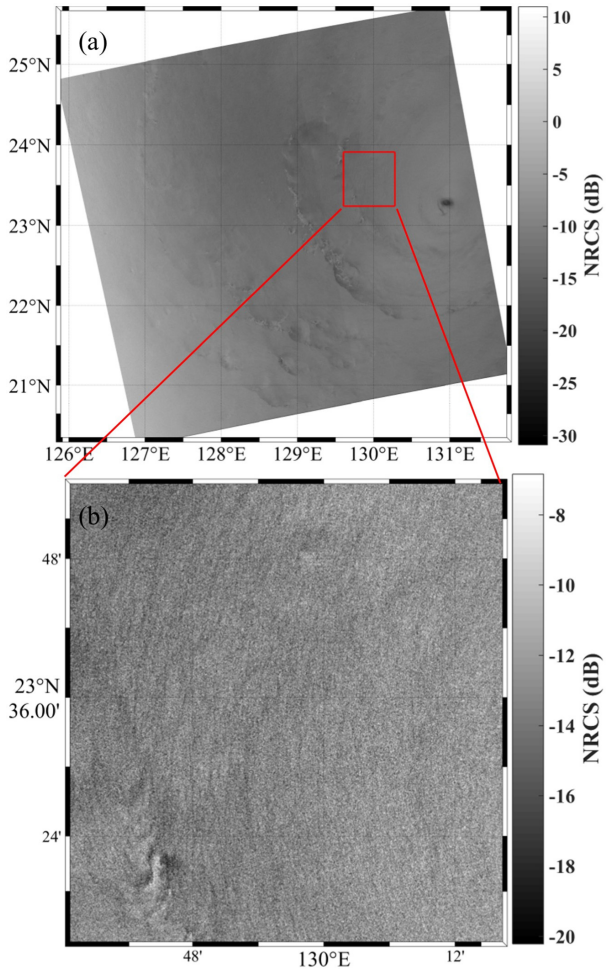


Fig. 1. SAR image of TC Lionrock acquired at 09:30 UTC on August 25, 2016 by RADARSAT-2. (a) Full-scene SAR image with a resolution of 50 m and coverage of 500 km × 500 km. (b) Enlargement of the red box in (a), where the stripe-like features are WIOTs.

determination techniques, such as wavelet analysis, watershed segmentation, morphology analysis, and a particle swarm optimization algorithm [22], [26]–[28], have been developed. Briefly, the algorithms for estimating TC center position from SAR imagery have only been developed in recent years. Most of the algorithms are semi-automatic and require human involvement. Moreover, due to the complexity of TC patterns in SAR imagery, the research on the automatic TC-center estimation algorithm with high robustness is still challenging.

In this article, we proposed an automatic TC-center estimation algorithm on the basis of all retrieved sea surface wind directions (SSWDs) in an entire SAR image, and a large number of SAR images (87 SAR images) are used to verify the algorithm. As far as we know, this is the first statistic study for automatically estimating TC center position. SSWD retrieval from SAR images is usually based on the wind-induced oriented texture (WIOT). Horizontal rolls (also called roll vortices) can be generated by either dynamic or thermodynamic instability in the marine atmospheric boundary layer [29]. These rolls can lead to the sea surface wind convergence and divergence, producing WIOTs (Fig. 1) with alternating brightness and darkness patterns in the SAR image. The WIOTs are generally aligned with SSWDs. Foster [30] showed that boundary layer rolls are widely distributed in TC systems. Thus, the SSWDs of a TC system can be estimated from WIOTs extracted from an SAR image. Several WIOT-based SSWD retrieval methods have been published, including the 2-D Fourier transform method [31]–[35], the local gradient method [36], [37], the improved local gradient (ILG) method [38], and the gray-level co-occurrence matrix-based texture orientation estimation method [39], [40]. In this study, we adopted the ILG method to retrieve TC SSWDs, and our SSWD-based TC-center estimation algorithm includes two main stages and one optional stage that can be summarized as follows.

Stage 1: Coarsely estimate the TC center position on the basis of the SSWDs retrieved from the whole SAR image. If it is located outside the SAR image, the coarsely estimated TC center position is the final result. Otherwise, trigger the following stage.

Stage 2: Extract the sub-SAR image (∼150 km × 150 km) centered at the coarsely estimated TC center position, and precisely estimate the TC center position from the sub-SAR image as done from the whole SAR image in Stage one.

Optional Stage: The above stages are only based on SSWD information. We can further adjust the precisely estimated center position, considering that normalized radar cross section (NRCS) is normally minimal at the TC center.

For clarity, we call the results of the first, second, and optional stages the coarsely estimated center position, the precisely estimated center position, and the NRCS-adjusted center position, respectively.

The remainder of this article is organized as follows. In Section II, we introduce our algorithm for estimating TC center positions. In Section III, we apply the algorithm to the 16 RADARSAT-2 and 71 Sentinel-1 SAR images and compare the estimated TC center positions with the visually located and best track (BT)-interpolated TC center positions. The discussion and summary are given in Section IV.

II. TC-Center Estimation Algorithm

A. Basic Principle of the Algorithm

If a TC's wind field has a circle pattern, the SSWD at each position will be aligned with the circumferential direction of the polar coordinate system centered at the TC center for the Northern Hemisphere (or aligned with the anti-circumferential direction for the Southern Hemisphere). However, the wind field patterns of TCs are spiral rather than circle, having SSWDs deviating inward from the circumferential direction. The deviation angles are called inflow angles in literature [41]. Due to the spiral pattern, the lines perpendicular to the SSWDs will not pass through the TC center unless the lines' inflow angles are compensated. However, the inflow angles are unknown for an SAR image before the TC center position is determined. Therefore, our algorithm sets several angles to compensate for the SSWD deviation and finds the best compensation angle having the maximum convergence degree of the perpendicular lines. The position that the perpendicular lines convergence under the best compensation angle is considered to be the TC center position.

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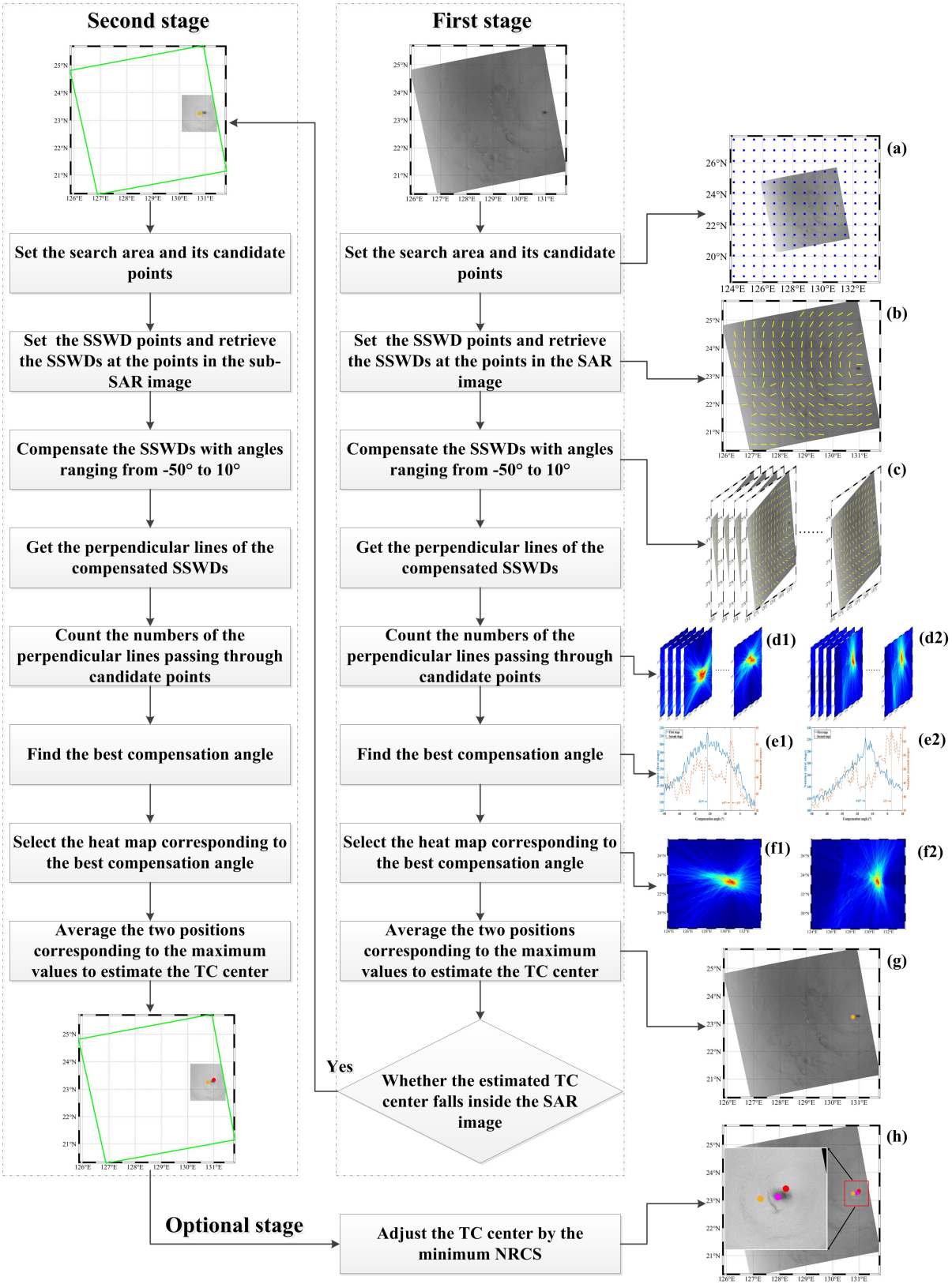


Fig. 2. Flowchart of the TC-center estimation algorithm. (a) Candidate points (141 × 149) are represented by the blue dots. (b) SSWDs are represented by the yellow lines. (c) SSWDs are deflected outward from −50° to 10° at an interval of 0.5° (the negative angles represent clockwise deflection). (d1) Heat maps are generated by the heat-map-generating method with the column-scanning. (d2) Heat maps are generated by the heat-map-generating method with the row-scanning. (e1) Maximum value in each heat map of (d1) is found, as shown in Fig. 5(a). (e2) Maximum value in each heat map of (d2) is found, as shown in Fig. 5(b). (f1) Heat map is selected from (d1) according to the best compensation angle, as shown in Fig. 6(a). (f2) Heat map is selected from (d2) according to the best compensation angle, as shown in Fig. 6(b). (g) Coarsely estimated center position is represented by the orange dot. (h) Coarsely estimated, precisely estimated, and NRCS-adjusted center position are represented by the orange, red, and magenta dot, respectively.

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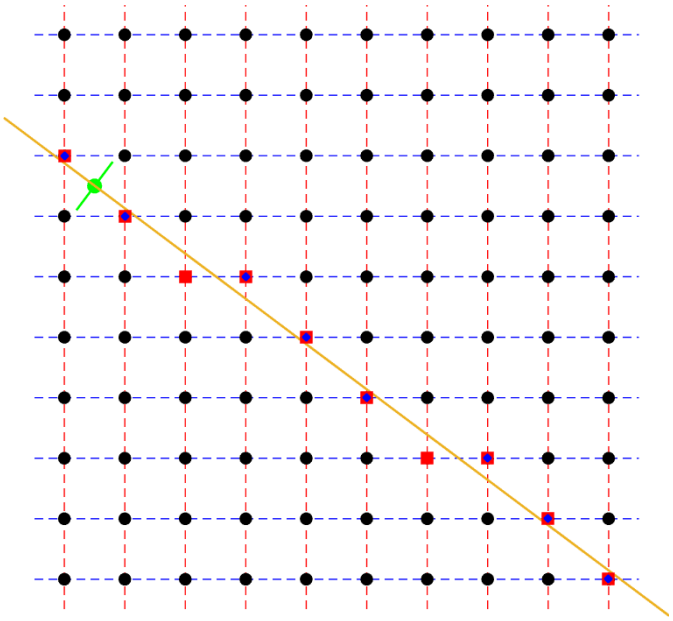


Fig. 3. Example of how to efficiently find the candidate points passed by the line perpendicular to a compensated SSWD. Candidate points and the SSWD point are denoted by the black and green dots, respectively. The compensated SSWD at the SSWD point is denoted by the green line. The compensated SSWD's perpendicular line is denoted by the orange line. The column and row lines are denoted by the red and blue dotted lines, respectively. The candidate points closest to each cross point of the orange line and column lines are considered to be passed by the orange line, and these candidate points are denoted by the red squares. The candidate points closest to each cross point of the orange line and row lines are also considered to be passed by the orange line, and these candidate points are denoted by the blue diamonds.

B. Location Procedure

The procedure of our TC-center estimation algorithm has two main stages and one optional stage. In the first stage, the TC center position is coarsely estimated from a TC SAR image. If the coarsely estimated center position is outside the TC SAR image, the procedure is stopped, and this position is the final result. Otherwise, the sub-SAR image (∼150 km × 150 km) centered at the coarsely estimated center position is extracted for the second stage to precisely estimate the center position. Optionally, the precisely estimated center position can be further adjusted by the minimum NRCS. For clarity, Fig. 2 shows a typical example (TC Lionrock-2016) to illustrate our algorithm. The row and column numbers (M, N) of the SAR image are 12 071 and 13305, and the resolution is 50 m.

Without loss of generality, the TC SAR image has $M \times N$ pixels. For the first stage of the algorithm, the TC center could be outside the SAR image. Thus, the search area is set to be co-centered on the SAR image but has a comparatively larger size of $(M + 2\ L) \times (N + 2L)$ pixels. Candidate points are set in each row and column of the search area at the P pixels interval [Fig. 2(a)]. If there are candidate points over land, then these candidate points are excluded. The TC center position will be determined from these candidate points.

Additionally, points are set in each row and column of the SAR images at the Q pixels interval, and for clarity, the points are called SSWD points. The SSWDs at the SSWD points are retrieved by the ILG method from the subimages

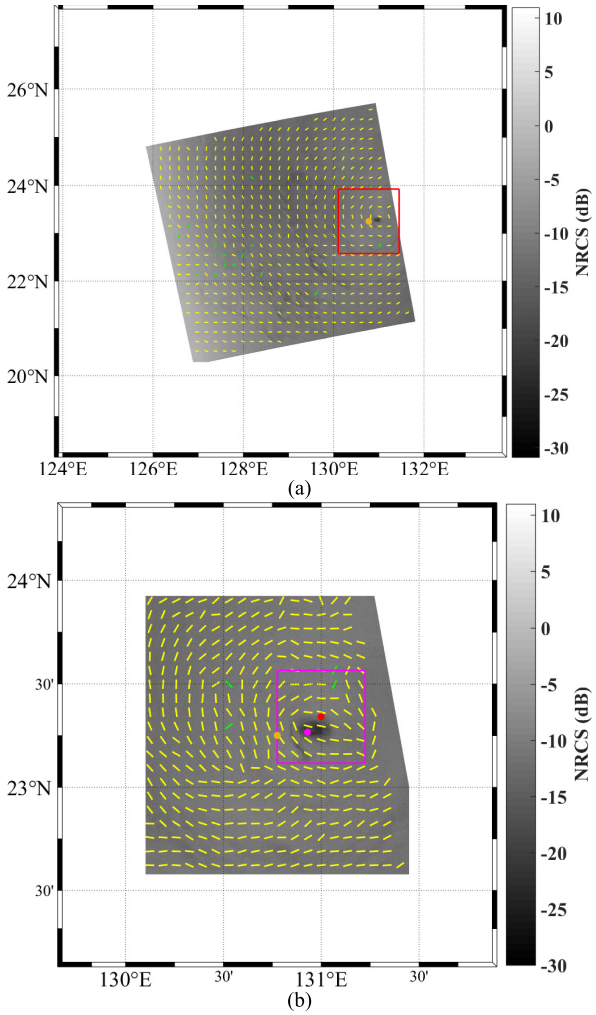


Fig. 4. SSWDs of TC Lionrock-2016 are retrieved by the ILG method. The removed SSWDs are denoted by the green lines, and the rest SSWDs are denoted by the yellow lines. (a) Coarsely estimated center position is denoted by the orange dot. The subimage inside the red box is centered at the orange dot and is used to precisely estimate the center position. (b) Precisely estimated center position is denoted by the red dot and is adjusted by the minimum NRCS from the subimage inside the magenta box. The NRCS-adjusted center position is denoted by the magenta dot. For comparison, the coarsely estimated center position is also denoted by the orange dot in (b).

($m \times m$ pixels) centered at the SSWD points [Fig. 2(b)]. However, the WIOTs are too weak to accurately retrieve the SSWDs in some regions of SAR images or are affected by possible interferences (e.g., beam seams). Therefore, we used the following method to remove SSWD outliers before locating the TC center position: slide a $(2A + 1) \times (2A + 1)$ window over SSWD points and for the SSWD point (i, j), calculate

$$S = \left(\sin 2\theta_{i,j} - \frac{\sum_{i'=-A, i' \neq i}^{A} \sum_{j'=-A, j' \neq j}^{A} \sin 2\theta_{i',j'}}{(2A+1)^2 - 1} \right)^2 + \left(\cos 2\theta_{i,j} - \frac{\sum_{i'=-A, i' \neq i}^{A} \sum_{j'=-A, j' \neq j}^{A} \cos 2\theta_{i',j'}}{(2A+1)^2 - 1} \right)^2 \quad (1)$$

$\theta_{i,j}$ represents the SSWD at the SSWD point (i, j). If $S >$ threshold, the SSWD at the SSWD point (i, j) is considered

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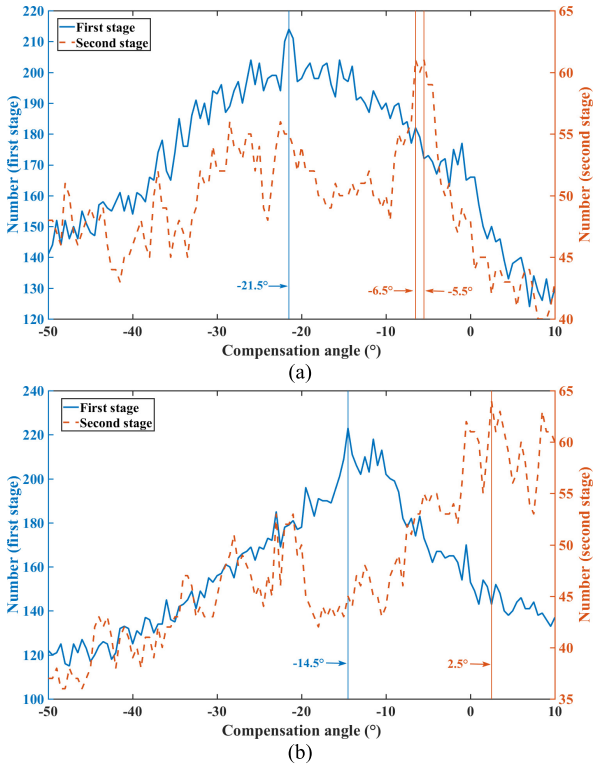


Fig. 5. Blue and orange dotted lines show the maximum values in each heat map of the first stage and second stage, respectively. The maximum values are obtained from the heat maps generated by the heat-map-generating method with (a) column-scanning and (b) row-scanning.

to be an outlier and is removed. The A is set to 2, and the threshold is set to 1.0 in the first stage and 1.5 in the second stage of the algorithm.

Overall, 121 compensation angles are set in the location procedure. They range from $-50°$ to $10°$ at an interval of $0.5°$. Here, the negative angles represent clockwise deflection for the Northern Hemisphere and anticlockwise deflection for the Southern Hemisphere. These compensation angles are used in sequence to deflect the retrieved SSWDs, as shown in Fig. 2(c).

To efficiently find the candidate points passed by the line perpendicular to a compensated SSWD, we developed a heat-map-generating method including two steps. First, the cross points of the perpendicular line and the column lines are located, and the candidate points closest to the cross points are found if the cross points are within the search area (Fig. 3). These candidate points are considered to be passed by the perpendicular line. For brevity, we call this search procedure column-scanning. Second, the heat map is generated by counting the numbers of perpendicular lines passing through the candidate points. Thus, there are 121 heat maps [Fig. 2(d1)] and the 121 maximum values [Fig. 2(e1)] in these heat maps for the 121 compensation angles. The heat map [Fig. 2(f1)] corresponding to the global maximum value in all heat maps is found, and the candidate point corresponding to the global maximum value is the estimated quasi-center position. In the case of the global maximum value occurring at different compensation angles, the candidate points' positions with the global maximum value in the maps are averaged to obtain the estimated quasi-center position. However, all the above outputs are on the basis of the column-scanning. The estimated quasi-center position will move slightly if the search procedure is changed to the following: the cross points of the perpendicular line and the row lines are located, and the candidate points closest to the cross points are found. For brevity, we call this search procedure the row-scanning. Similarly, the 121 heat maps [Fig. 2(d2)] are also generated for 121 compensation angles, and the 121 maximum values are also found [Fig. 2(e2)]. The heat map [Fig. 2(f2)] corresponding to the global maximum value in these heat maps is obtained. The candidate point corresponding to the global maximum value is the other estimated quasi-center position. Finally, the two estimated quasi-center positions are averaged to get the coarsely estimated TC center position [Fig. 2(g)].

The procedure in the second stage is the same as that in the first stage. The precisely estimated center position is obtained via the second stage. Optionally, the precisely estimated center position can be further adjusted from the sub-SAR image of its nearby area ($\sim$50 km $\times$ 50 km) by considering that NRCS is normally minimal at the TC center. The parameters L, P, Q, and m can be adjusted for different resolutions. For the SAR images with the resolutions of 40 and 50 m, the parameter values are set to 4500, 150, 150, and 400 in the first stage and 1000, 100, 100, and 400 in the second stage. For the SAR images with the resolution of 500 m, due to that the pixel interval is increased by ten times in comparison with the SAR images of 50 m resolution, the parameter values are correspondingly reduced by ten times. Consequently, the parameter values for 500 m resolution are set to 450, 15, 15, and 40 in the first stage and 100, 10, 10, and 40 in the second stage.

The SAR image of TC Lionrock-2016 is used to illustrate the entire procedure of the algorithm, as shown in Fig. 4(a) with the retrieved SSWDs. In the first stage, the 121 maximum values of heat maps generated with the column- and row-scannings are denoted by the blue line in Fig. 5(a) and (b), respectively. The blue line's maximum values occur at the compensation angle of $-21.5°$ and $-14.5°$. The heat maps corresponding to the two compensation angles are shown in Fig. 6(a) and (b), respectively. The candidate points corresponding to the maximum values in Fig. 6(a) and (b) are found as the coarsely estimated quasi-center positions. The two quasi-center positions are averaged to get the coarsely estimated TC center position and are denoted by the orange dot in Fig. 4(a) and (b). The subimage centered at the orange dot and denoted by the red box in Fig. 4(a) is extracted for the second stage. The subimage and the retrieved SSWDs are shown in Fig. 4(b). In the second stage, the 121 maximum values of the heat maps generated with the column- and row-scannings are denoted by the orange dotted line in Fig. 5(a) and (b), and the heat maps corresponding to the compensation angles of $-6.5°$ and $2.5°$ are given in Fig. 6(c) and (d), respectively. The two maximum values in the heat maps corresponding

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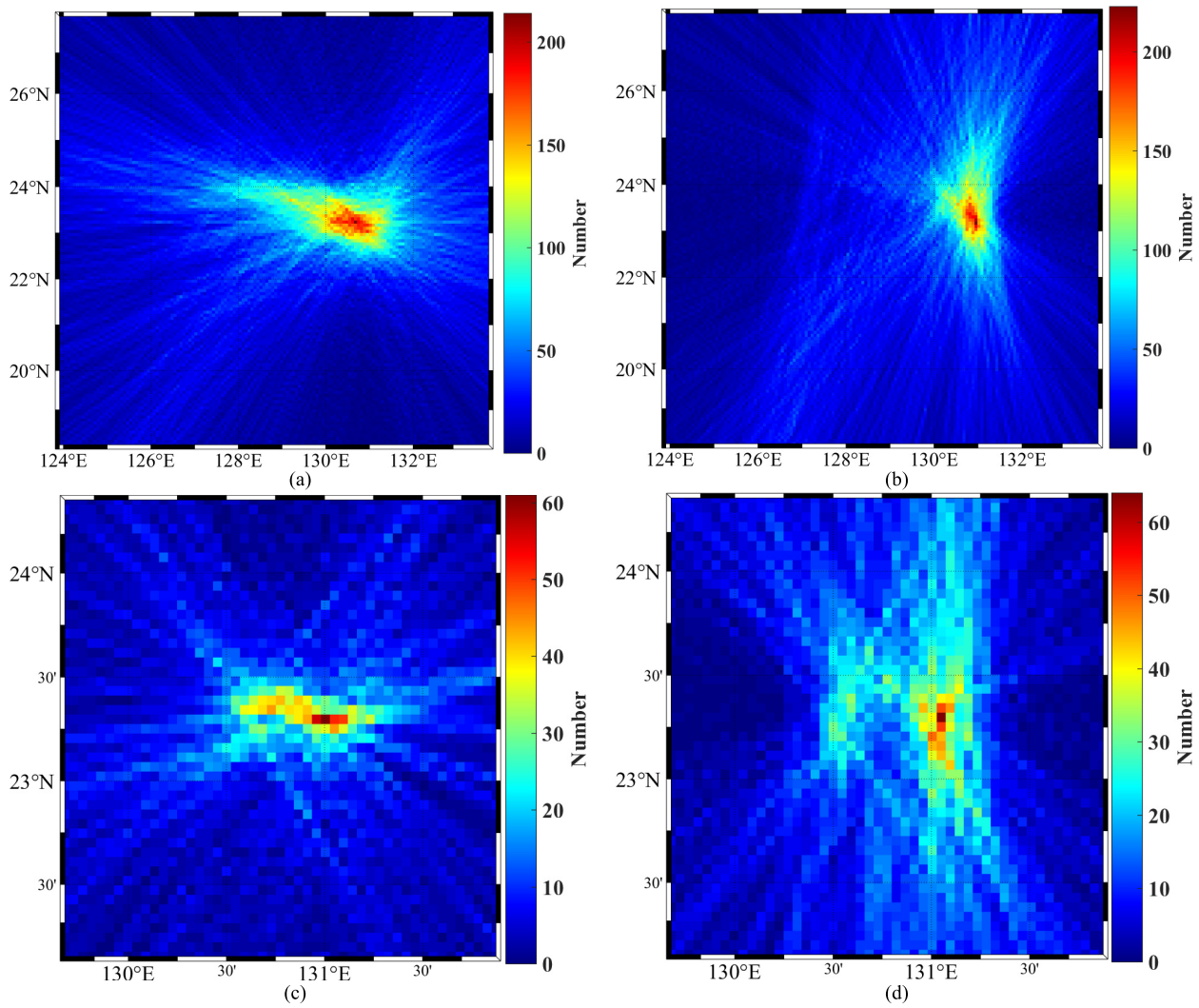


Fig. 6. Heat maps generated by counting the numbers of the compensated SSWDs' perpendicular lines passing through each candidate point. (a) Heat map corresponding to the compensation angle of −21.5° is generated with the column-scanning in the first stage. (b) Heat map corresponding to the compensation angle of −14.5° is generated with the row-scanning in the first stage. (c) Heat map corresponding to the compensation angle of −6.5° is generated with the column-scanning in the second stage. (d) Heat map corresponding to the compensation angle of 2.5° is generated with the row-scanning in the second stage.

to the compensation angle of −6.5° and −5.5° are found, and the two corresponding candidate points are averaged to get the precisely estimated quasi-center position. The candidate point corresponding to the maximum value in Fig. 6(d) is found as the other precisely estimated quasi-center position. The two quasi-center positions are averaged to get the precisely estimated TC center position denoted by the red dot, and the NRCS-adjusted center position is denoted by the magenta dot in Fig. 4(b).

III. Validation

A. Data

This study collected 87 SAR images, as listed in Tables I and II.

In Table I, the 16 RADARSAT-2 SAR images are acquired in ScanSAR wide mode (SCW) with a resolution of 50 m, and the 14 Sentinel-1 SAR images were acquired in interferometric wide swath mode (IW) and extra-wide swath mode (EW) with a resampling resolution of 40 m. The 30 SAR images, shown in Figs. 7 and 8, are vertical transmit and vertical receive (VV polarization) data. The TC centers are inside the 25 SAR images shown in Fig. 7 but outside the five SAR images shown in Fig. 8. In Table II, the 57 Sentinel-1 SAR images were also acquired in IW mode and EW mode but with a resolution of 500 m. The 57 SAR images containing TC centers are vertical transmit and horizontal receive (VH polarization) data in Table II. Besides speckle noise, the along- and cross-track beam seams in the SAR images (more apparent in VH polarization) could also lead to the outliers in the retrieved SSWDs.

BT datasets were collected. For the Western North Pacific, we used the BT dataset provided by the China Meteorological Administration (CMA) TC Data Center [42]. For the Eastern Pacific and the Atlantic, we used the BT dataset provided by the National Oceanic and Atmospheric Administration (NOAA) [43]. TC center position records are given every 6 h, and we temporally interpolated the data at SAR imaging times.

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TABLE I

TC CENTER POSITIONS ESTIMATED FROM THE SAR IMAGES AND THOSE FROM THE VISUALLY LOCATED AND BT-INTERPOLATED DATA. FOR THE SAR IMAGES CONTAINING THE TC CENTERS, THE VISUALLY LOCATED TC CENTER POSITIONS ARE USED AS THE GROUND TRUTH DATA. FOR THE SAR IMAGES NOT CONTAINING THE TC CENTERS, THE TC CENTER POSITIONS CANNOT BE VISUALLY LOCATED, AND BT-INTERPOLATED CENTER POSITIONS ARE USED AS THE GROUND TRUTH DATA. D_1 REPRESENTS THE DIFFERENCES BETWEEN THE COARSELY ESTIMATED CENTER POSITIONS AND THE GROUND TRUTH DATA. D_2 REPRESENTS THE DIFFERENCES BETWEEN THE PRECISELY ESTIMATED CENTER POSITIONS AND THE GROUND TRUTH DATA. D_3 REPRESENTS THE DIFFERENCES BETWEEN THE NRCS-ADJUSTED CENTER POSITIONS AND THE GROUND TRUTH DATA. THE OTHER INFORMATION GIVEN HERE INCLUDES TC NAME, IMAGING TIME, SAR NAME, MODE, SWATH, AND RESOLUTION AFTER RESAMPLING

No.	Name	UTC time		SAR				Visually-located					BT		
		dd/mm/yyyy	hh:mm	SAR	Resolution (m)	Swath (km)	Mode	Lat (°)	Lon (°)	D_1 (km)	D_2 (km)	D_3 (km)	Lat (°)	Lon (°)	D_1 (km)
01	Cristina	15/06/2014	13:23	R2	50	500	SCW	20.1 N	113.2 W	2.2	4.4	11.5	-	-	-
02	Arthur	03/07/2014	11:14	R2	50	500	SCW	31.6 N	78.7 W	19.1	10.5	6.7	-	-	-
03	Iselle	07/08/2014	15:58	R2	50	500	SCW	18.6 N	151.0 W	36.6	30.3	8.7	-	-	-
04	Karina	14/08/2014	01:47	R2	50	500	SCW	17.0 N	113.6 W	9.7	8.5	4.2	-	-	-
05	Karina	17/08/2014	14:26	R2	50	500	SCW	17.6 N	128.2 W	7.5	20.0	36.4	-	-	-
06	Norbert	07/09/2014	01:50	R2	50	500	SCW	25.5 N	115.4 W	7.0	6.2	3.7	-	-	-
07	Simon	03/10/2014	13:14	R2	50	500	SCW	18.4 N	109.5 W	11.8	15.9	42.5	-	-	-
08	Vance	01/11/2014	12:30	R2	50	500	SCW	9.3 N	102.8 W	4.5	5.8	6.4	-	-	-
09	Vance	02/11/2014	01:12	R2	50	500	SCW	10.2 N	104.9 W	20.6	17.3	51.8	-	-	-
10	Soudelor	01/08/2015	19:50	R2	50	500	SCW	14.5 N	148.9 E	56.3	13.6	1.2	-	-	-
11	Soudelor	03/08/2015	20:31	R2	50	500	SCW	17.9 N	140.2 E	40.0	17.3	31.6	-	-	-
12	Lionrock	25/08/2016	09:30	R2	50	500	SCW	23.2 N	130.9 E	22.9	8.3	5.6	-	-	-
13	Meranti	10/09/2016	20:48	R2	50	500	SCW	16.1 N	136.5 E	55.1	18.9	14.9	-	-	-
14	Haima	15/10/2016	20:29	R2	50	500	SCW	9.9 N	139.9 E	1.4	2.4	2.2	-	-	-
15	Noru	30/07/2017	20:25	R2	50	500	SCW	22.7 N	140.7 E	21.2	21.5	7.4	-	-	-
16	Lester	30/08/2016	14:46	S1A	40	400	EW	17.9 N	134.3 W	45.1	10.0	8.3	-	-	-
17	Lester	31/08/2016	03:15	S1A	40	400	EW	17.6 N	136.5 W	18.2	13.4	4.2	-	-	-
18	Gaston	01/09/2016	20:30	S1A	40	400	EW	38.1 N	38.4 W	45.4	79.5	60.5	-	-	-
19	Hermine	01/09/2016	23:45	S1A	40	250	IW	29.1 N	84.7 W	6.5	2.7	7.3	-	-	-
20	Namtheun	04/09/2016	09:21	S1A	40	400	EW	32.0 N	129.5 E	4.8	9.6	7.5	-	-	-
21	Karl	23/09/ 2016	22:23	S1A	40	400	EW	29.9 N	65.1 W	30.4	4.8	10.5	-	-	-
22	Irma	07/09/2017	10:30	S1A	40	250	IW	20.0 N	68.7 W	23.8	20.1	6.8	-	-	-
23	Jose	08/09/2017	22:03	S1A	40	250	IW	16.6 N	58.5 W	3.1	11.5	1.5	-	-	-
24	Maria	18/09/2017	09:50	S1A	40	250	IW	14.3 N	59.5 W	5.4	7.0	3.6	-	-	-
25	Maria	21/09/2017	22:45	S1A	40	250	IW	20.7 N	70.0 W	39.1	37.1	17.3	-	-	-
26	Norbert	07/09/2014	01:48	R2	50	500	SCW	-	-	-	-	-	25.1 N	115.4 W	40.5
27	Hermine	01/09/2016	23:44	S1A	40	250	IW	-	-	-	-	-	28.6 N	84.7 W	45.7
28	Noru	05/08/2017	09:28	S1A	40	250	IW	-	-	-	-	-	30.0 N	129.9 E	47.3
29	Maria	21/09/2017	22:46	S1A	40	250	IW	-	-	-	-	-	20.5 N	70.0 W	82.9
30	Saola	28/10/2017	09:28	S1A	40	250	IW	-	-	-	-	-	27.9 N	128.9 E	5.3

TC does not move at a uniform velocity in the BT record interval of 6 h. The visually located and BT-interpolated TC center positions were used as the ground truth for the images containing and not containing TC centers, respectively.

B. Results

All the coarsely estimated, precisely estimated, and NRCS-adjusted center positions of Table I (Nos. 1–25) are shown with the visually located center positions in Fig. 7; these features are denoted by orange, red, magenta, and green dots. The coarsely estimated center positions of Table I (Nos. 26–30) are shown with the BT-interpolated center positions in Fig. 8; these features are denoted by orange and black dots. Fig. 7 shows that in most cases, compared with the coarsely estimated center positions, the precisely estimated center positions are closer to the visually located TC center

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TABLE II

TC CENTER POSITIONS ESTIMATED FROM THE SENTINEL-1 SAR IMAGES. THE MEANING OF D_1, D_2, AND D_3 ARE THE SAME AS THOSE IN TABLE I. THE OTHER INFORMATION GIVEN HERE INCLUDES TC NAME, IMAGING TIME, SAR NAME, AND MODE. FURTHERMORE, THE 57 SAR IMAGES ARE VH POLARIZATION DATA WITH THE RESOLUTION OF 500 m.

Name	UTC time		SAR		D_1 (km)	D_2 (km)	D_3 (km)	Name	UTC time		SAR		D_1 (km)	D_2 (km)	D_3 (km)
	dd/mm/yyyy	hh:mm	SAR	Mode					dd/mm/yyyy	hh:mm	SAR	Mode			
Jongdari	24/07/2018	20:55	S1A	EW	4.0	6.1	3.6	Dorian	30/08/2019	22:46	S1A	IW	4.3	5.0	27.7
Shanshan	07/08/2018	08:32	S1A	EW	5.0	1.6	23.0	Juliette	04/09/2019	13:40	S1A	EW	14.5	5.4	3.5
Hector	07/08/2018	15:44	S1A	EW	2.1	2.5	2.1	Lerena	20/09/2019	13:06	S1A	IW	7.6	4.7	4.9
Shanshan	07/08/2018	20:35	S1A	IW	4.5	4.6	4.0	Hagibis	11/10/2019	20:53	S1A	EW	32.0	16.2	8.8
Hector	08/08/2018	04:13	S1A	EW	6.0	6.3	4.3	Halong	05/11/2019	19:58	S1A	EW	3.4	2.2	3.6
Cimaron	22/08/2018	21:01	S1A	IW	45.7	34.5	36.4	Halong	07/11/2019	19:40	S1A	EW	19.2	11.5	2.5
Lane	23/08/2018	04:38	S1A	IW	12.5	5.3	6.0	Bulbul	09/11/2019	12:12	S1A	IW	2.7	4.6	11.0
Mariam	31/08/2018	03:32	S1A	EW	3.9	3.2	12.7	Herold	16/03/2020	02:01	S1A	EW	10.3	3.0	10.1
Jebi	31/08/2018	20:39	S1A	EW	4.9	2.5	1.6	Jongdari	25/07/2018	20:46	S1B	EW	16.8	6.1	2.1
Norman	05/09/2018	03:41	S1A	EW	15.5	8.1	22.2	Jongdari	27/07/2018	08:22	S1B	EW	27.9	5.9	1.5
Florence	08/09/2018	09:39	S1A	EW	23.0	2.1	1.6	Hector	10/08/2018	16:57	S1B	EW	14.7	13.7	2.7
Mangkhut	14/09/2018	09:51	S1A	EW	10.0	5.6	6.7	Soulik	18/08/2018	08:38	S1B	EW	9.7	10.3	6.8
Trami	25/09/2018	21:20	S1A	EW	14.5	4.6	3.8	Soulik	18/08/2018	20:45	S1B	EW	12.4	3.9	2.8
Rosa	29/09/2018	02:02	S1A	EW	14.6	4.0	0.7	Jebi	29/08/2018	07:55	S1B	EW	44.8	2.0	2.9
Kong-rey	02/10/2018	21:12	S1A	EW	10.3	4.0	3.0	Norman	02/09/2018	14:37	S1B	EW	5.6	9.1	10.1
Sergio	03/10/2018	13:41	S1A	EW	14.9	8.5	7.8	Norman	07/09/2018	04:14	S1B	EW	23.6	12.6	7.8
Sergio	04/10/2018	02:09	S1A	EW	33.1	4.1	9.6	Mangkhut	11/09/2018	20:48	S1B	EW	20.2	3.2	5.0
Sergio	06/10/2018	14:06	S1A	EW	18.5	4.5	8.5	Rosa	30/09/2018	01:54	S1B	EW	7.0	4.5	7.9
Sergio	07/10/2018	02:34	S1A	EW	3.2	5.9	7.2	Haleh	07/03/2019	00:50	S1B	EW	7.0	9.0	13.9
Michael	09/10/2018	23:44	S1A	EW	27.4	16.9	11.0	Idai	14/03/2019	16:06	S1B	IW	7.2	5.2	5.1
Michael	10/10/2018	11:50	S1A	EW	23.2	4.0	2.1	Trevor	21/03/2019	20:15	S1B	EW	15.2	5.4	11.2
Leslie	13/10/2018	07:17	S1A	EW	56.8	38.6	9.1	Veronica	23/03/2019	21:39	S1B	IW	9.9	7.9	16.2
Yutu	25/10/2018	20:32	S1A	EW	5.4	5.6	4.3	Fani	03/05/2019	00:13	S1B	EW	25.1	6.3	0.0
Yutu	26/10/2018	09:01	S1A	EW	19.6	10.3	8.2	Vayu	14/06/2019	13:33	S1B	EW	13.9	8.4	2.2
Gelena	07/02/2019	02:09	S1A	EW	10.1	8.1	2.1	Barbara	04/07/2019	02:34	S1B	EW	39.8	8.1	8.5
Oma	14/02/2019	18:29	S1A	EW	18.8	9.0	18.9	Dorian	29/08/2019	10:21	S1B	IW	17.5	2.0	4.3
Idai	11/03/2019	02:43	S1A	EW	10.0	3.2	7.8	Halong	06/11/2019	19:48	S1B	EW	22.1	5.4	3.6
Veronica	22/03/2019	21:47	S1A	EW	22.3	4.9	1.6	Kaimaegi	19/11/2019	09:59	S1B	IW	44.3	14.8	9.3
Veronica	24/03/2019	10:38	S1A	EW	16.7	8.1	3.5								

positions. In Fig. 8, the SAR images do not contain TC centers. Even in these challenging cases, our algorithm still locates the TC center positions outside the SAR images with only a slightly decreased accuracy.

Figs. 9 and 10 clearly show the relative position relationship among all TC center positions. The two black diamonds in each panel represent the TC center positions reported in the BT dataset at times just before and after

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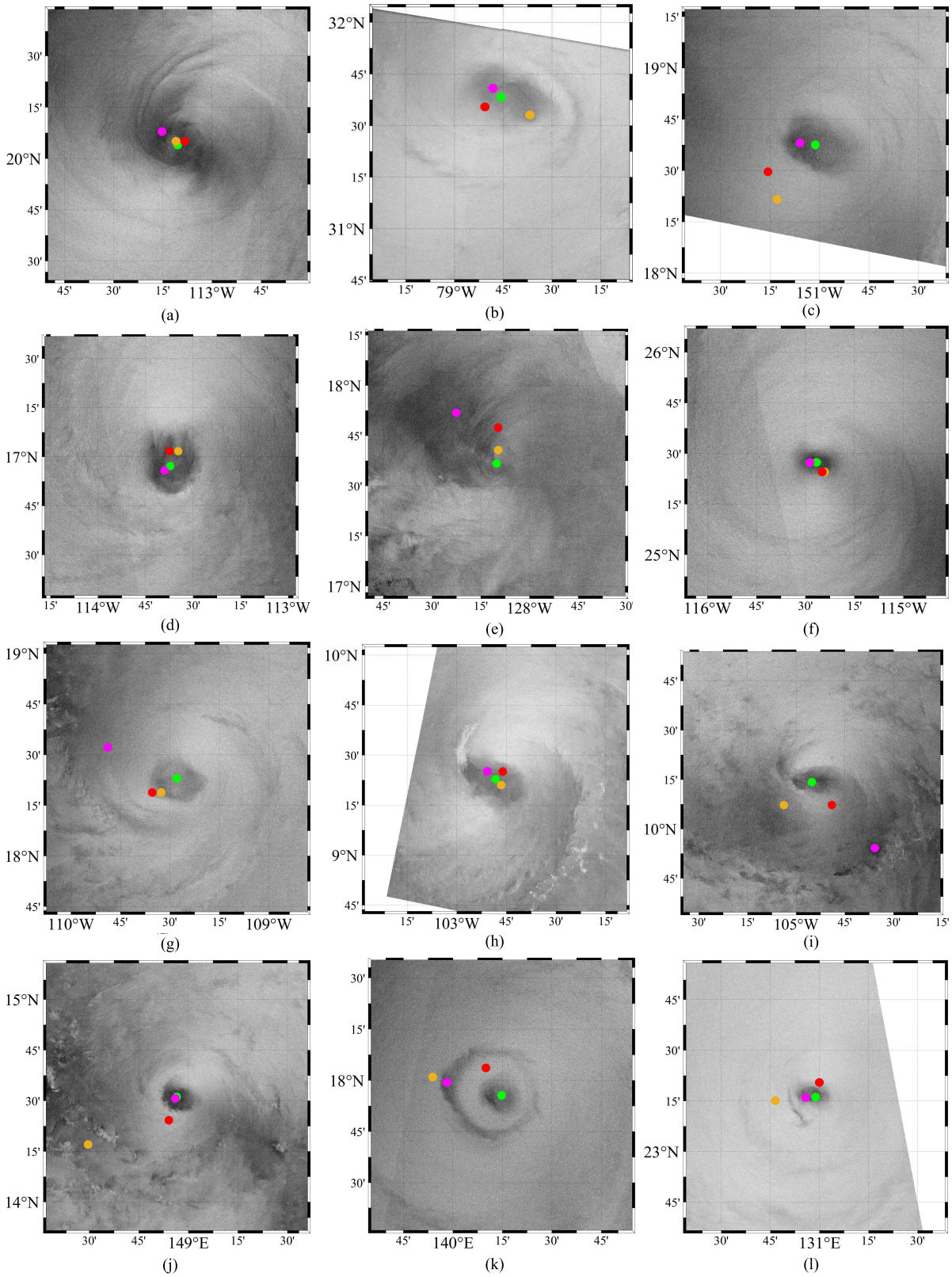


Fig. 7. Twenty-five SAR images containing TC centers were used to test our algorithm. The TC center areas were enlarged. The coarsely estimated, precisely estimated, NRCS-adjusted, and visually located center positions are represented by orange, red, magenta, and green dots, respectively. (a)–(y) TCs corresponding to Nos. 1–25 of Table I.

the imaging times, synthesized TC centers from post-event analysis [22].

For concise, let D_1 represent the differences between the coarsely estimated center positions and the ground truth data (the visually located and BT-interpolated TC center positions were used as the ground truth for the images containing and not containing TC centers, respectively), D_2 represents the differences between the precisely estimated center position

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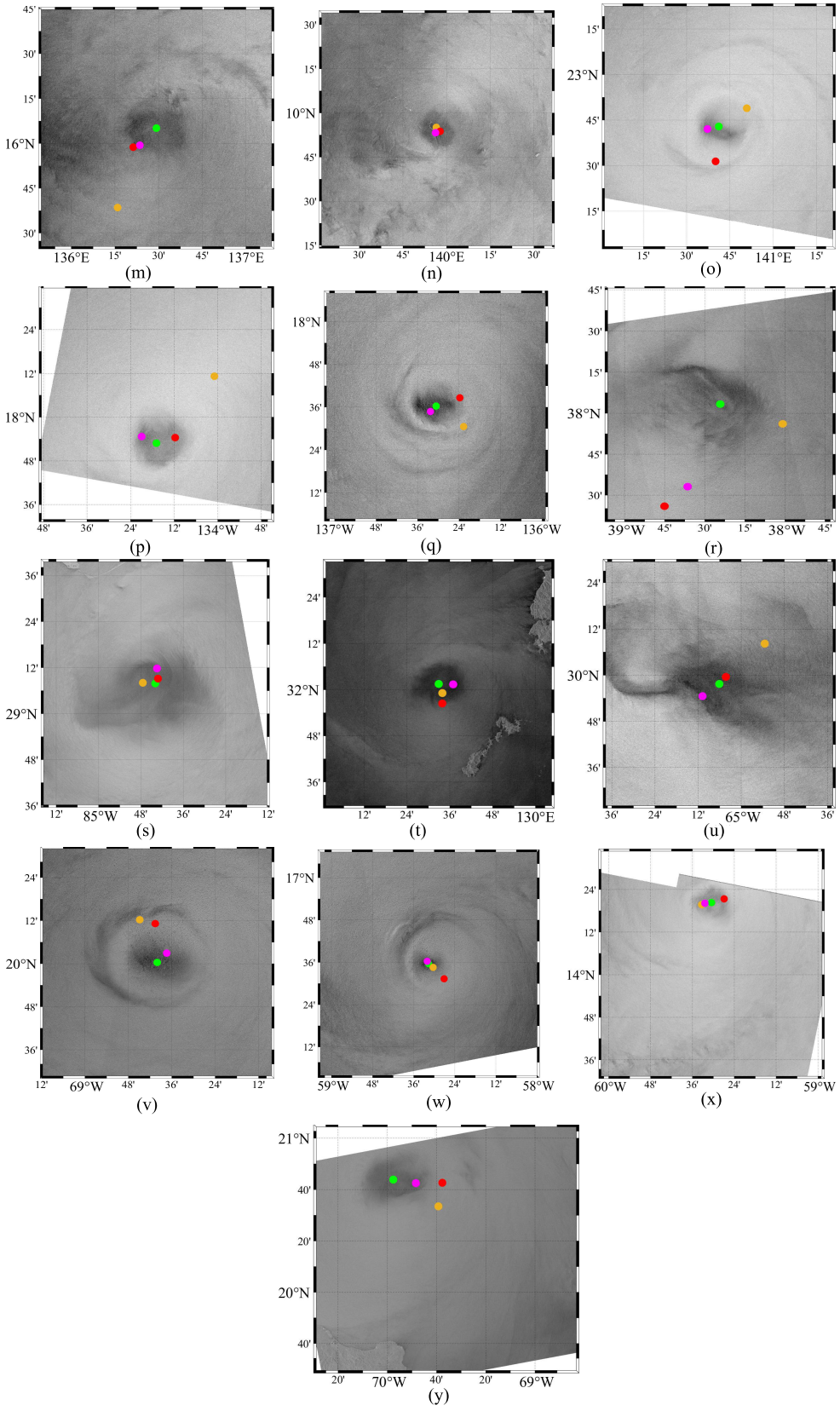


Fig. 7. *(Continued.)* Twenty-five SAR images containing TC centers were used to test our algorithm. The TC center areas were enlarged. The coarsely estimated, precisely estimated, NRCS-adjusted, and visually located center positions are represented by orange, red, magenta, and green dots, respectively. (a)–(y) TCs corresponding to Nos. 1–25 of Table I.

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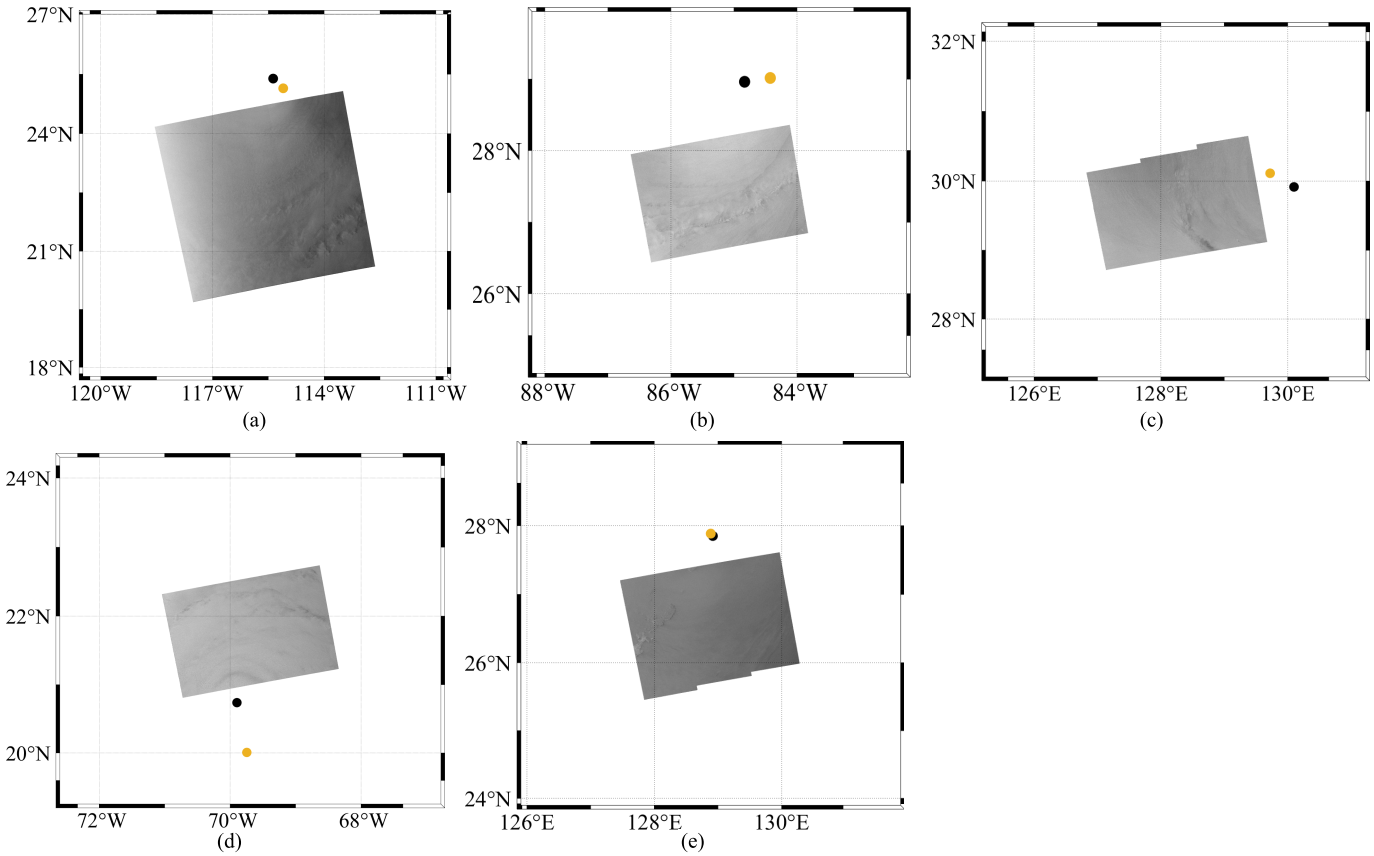


Fig. 8. Five SAR images not containing TC centers were used to test our algorithm. The coarsely estimated and BT-interpolated center positions are represented by orange and black dots, respectively. (a)–(e) TCs corresponding to Nos. 26–30 of Table I.

and the ground truth data, and D_3 represents the differences between the NRCS-adjusted center positions and the ground truth data. D_1–D_3 are shown in Tables I and II. For the 82 images containing TC centers, D_1–D_3 range from 1.4 to 56.8 km, 1.6 to 79.5 km, and 0.0 to 60.5 km, respectively, and the root mean squares of D_1–D_3 ($RMSD_1$, $RMSD_2$, and $RMSD_3$) are 22.8, 14.7, and 14.7 km. For the five images not containing TC centers, D_1 ranges from 5.3 to 82.9 km, and the $RMSD_1$ is 50.7 km. For all 87 SAR images, the $RMSD_1$ is 25.2 km.

IV. Discussion and Summary

In this study, we proposed a TC-center estimation algorithm for SAR imagery based on the fact that a straight line perpendicular to an SSWD after deflection with the compensation angle will pass through the TC center.

Our algorithm's procedure includes the first stage, the second stage, and the optional stage. In the first stage, the TC center position is coarsely estimated based on the SSWDs retrieved from an entire SAR image, and the coarsely estimated center position is the final result if it is located outside the SAR image. In the second stage, the sub-SAR image centered at the coarsely estimated center position is extracted, and the first-stage procedure is repeated with the sub-SAR image to precisely estimate the center position. In the optional stage, the precisely estimated center position is adjusted by considering that NRCS is normally minimal at the TC center.

We applied the algorithm to 87 SAR images, including five SAR images not containing TC centers. The algorithm-estimated TC center positions were compared with the visually located TC center positions and the temporally interpolated BT center positions. For the 82 images containing TC centers, the $RMSD_1$, $RMSD_2$, and $RMSD_3$ are 22.8, 14.7, and 14.7 km, respectively. For the rest five images not containing TC centers, the $RMSD_1$ is 50.7 km. For all 87 SAR images, the $RMSD_1$ is 25.2 km. Our algorithm achieved high accuracy.

However, our algorithm is on the basis of the TCs' SSWD pattern, so the decrease of the SSWD retrieval accuracy and the deformation of the SSWD pattern will degrade the algorithm's performance. As shown in Fig. 11, the geolocation of TC Gaston's SAR image [Fig. 7(r), 01/09/2016 20:30 universal time coordinated (UTC)] is represented by the green box, and $D_1 = 45.4$ km, $D_2 = 79.5$ km, and $D_3 = 60.5$ km. The geolocation of TC Leslie's SAR image (Fig. 12, 13/10/2018 07:17 UTC) is represented by the black box, and $D_1 = 56.8$ km, $D_2 = 38.6$ km, and $D_3 = 9.1$ km. The geolocation of TC Maria's SAR image (Fig. 7(y), 21/09/2017 22:45 UTC) is represented by the red box, and $D_1 = 39.1$ km, $D_2 = 37.1$ km, and $D_3 = 17.3$ km. The geolocation of the other TC Maria's SAR image (Fig. 8(d), 21/09/2017 22:46 UTC) is

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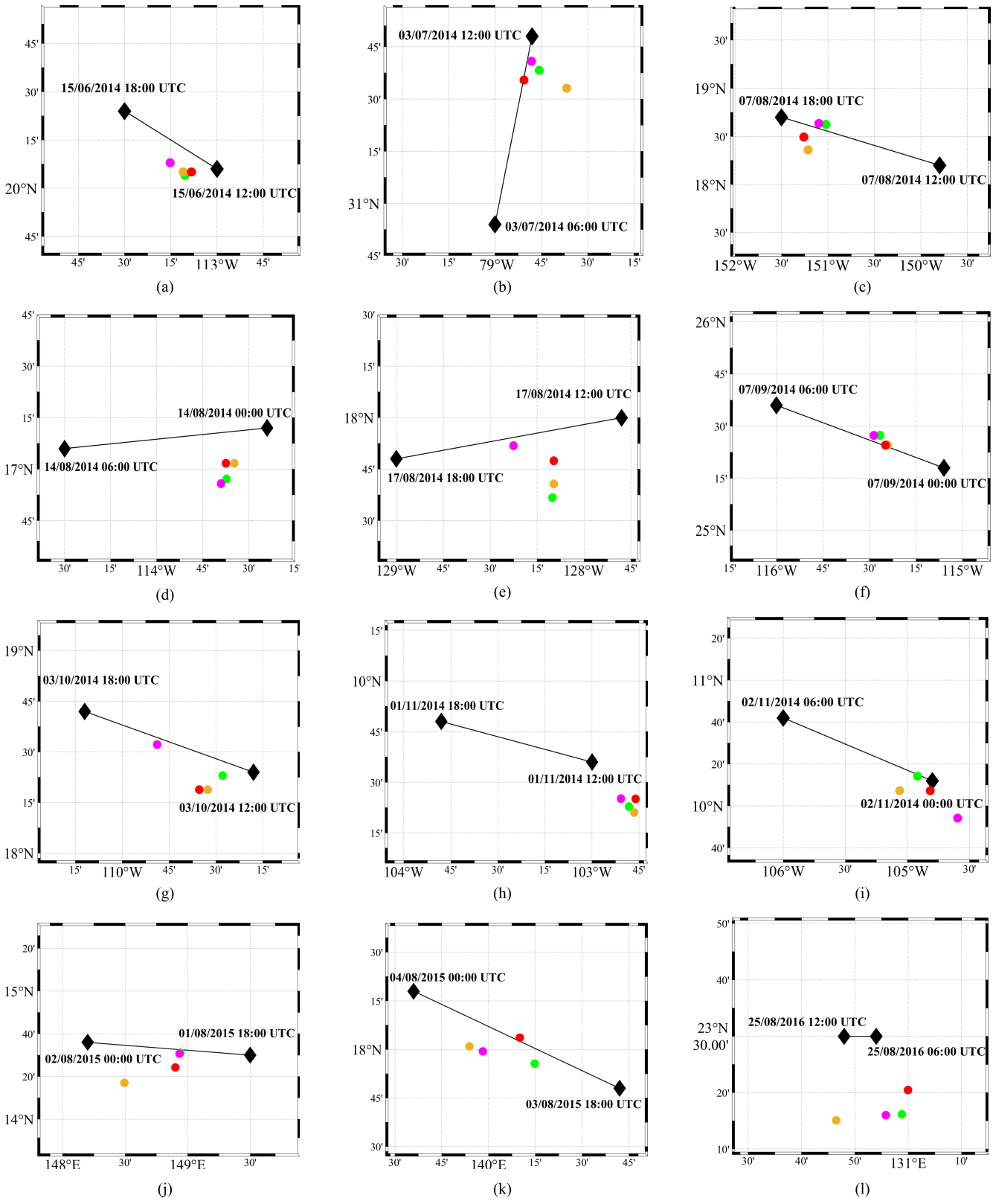


Fig. 9. Geolocations of the TC center positions in Fig. 7. The two black diamonds in each panel denote the TC center positions reported in the BT dataset at times just before and after the SAR imaging times, and the black line is the TC migration track suggested by the two diamonds. The coarsely estimated, precisely estimated, NRCS-adjusted, and visually located center positions are represented by orange, red, magenta, and green dots, respectively. (a)–(y) TCs corresponding to Nos. 1–25 of Table I.

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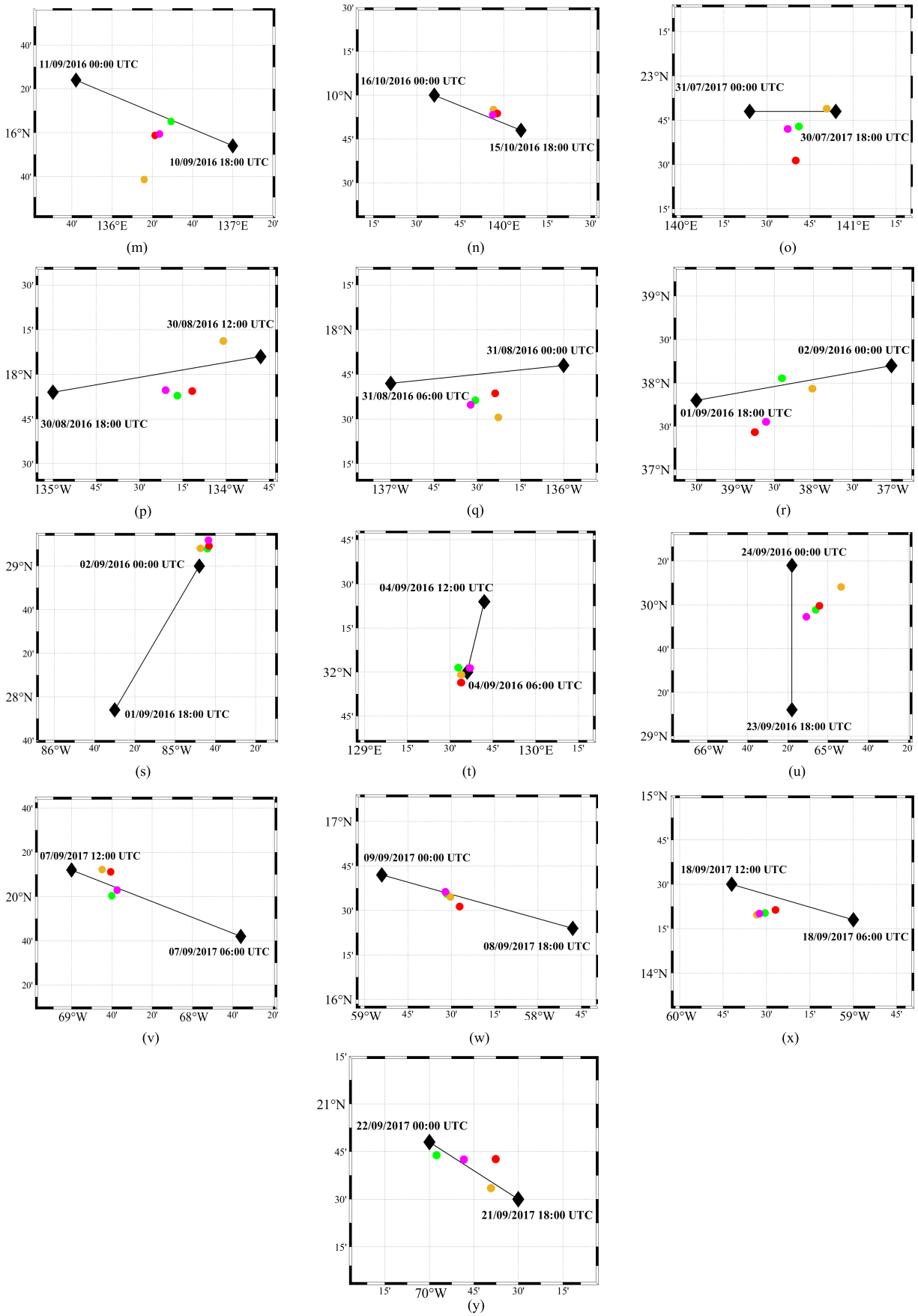


Fig. 9. *(Continued.)* Geolocations of the TC center positions in Fig. 7. The two black diamonds in each panel denote the TC center positions reported in the BT dataset at times just before and after the SAR imaging times, and the black line is the TC migration track suggested by the two diamonds. The coarsely estimated, precisely estimated, NRCS-adjusted, and visually located center positions are represented by orange, red, magenta, and green dots, respectively. (a)–(y) TCs corresponding to Nos. 1–25 of Table I.

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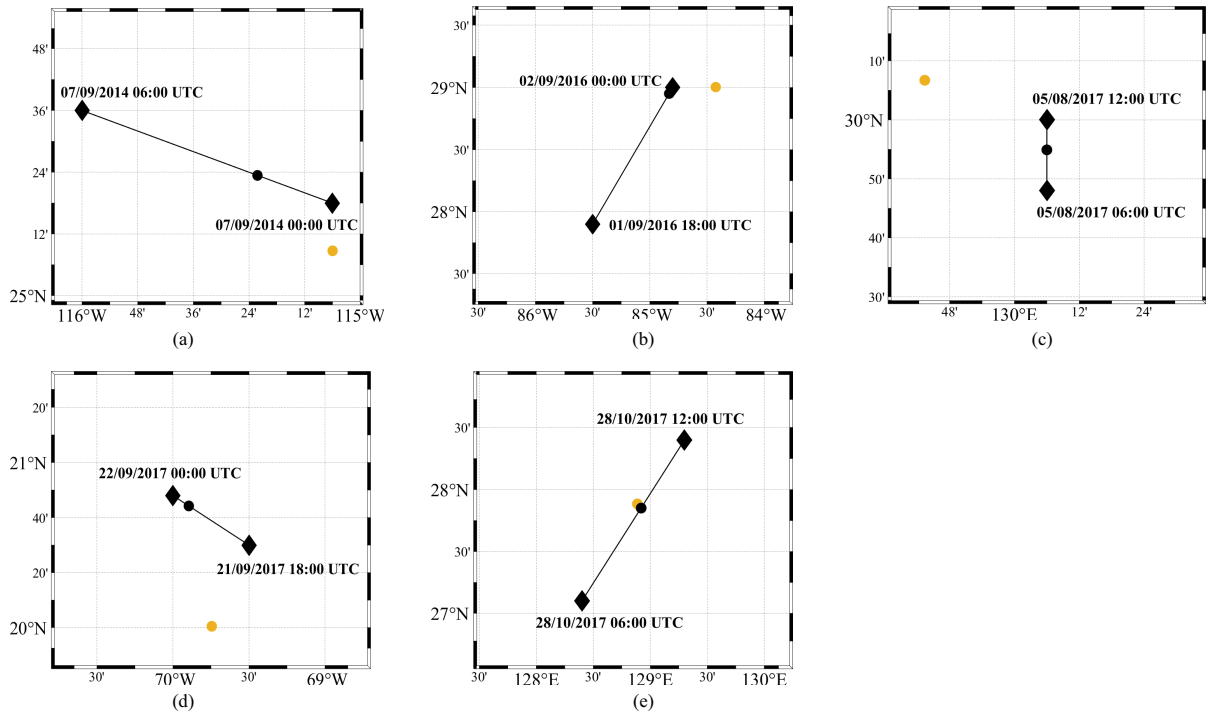


Fig. 10. Geolocations of the TC center positions in Fig. 8. The two black diamonds in each panel denote the TC center positions reported in the BT dataset at times just before and after the SAR imaging times, and the black line is the TC migration track suggested by the two diamonds. The coarsely estimated and BT-interpolated center positions are represented by orange and black dots, respectively. (a)–(e) TCs corresponding to Nos. 26–30 of Table I.

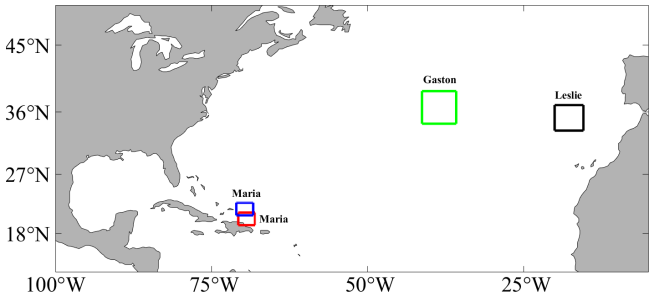


Fig. 11. Locations of the four SAR images corresponding to TCs Gaston, Maria, and Leslie.

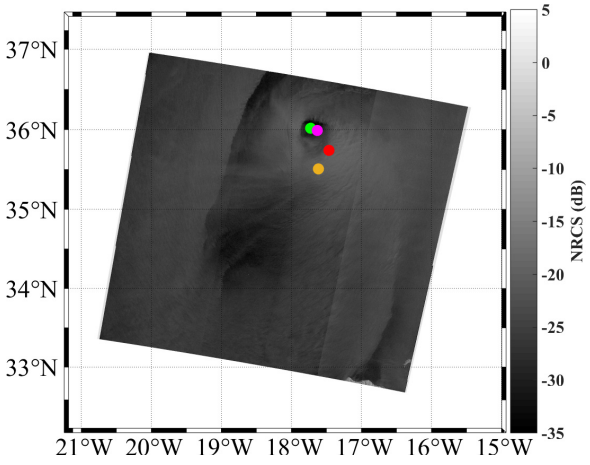


Fig. 12. Estimated center positions of TC Leslie. The coarsely estimated, precisely estimated, NRCS-adjusted, and visually located center positions are represented by orange, red, magenta, and green dots, respectively.

represented by the blue box. Since Fig. 8(d) is an SAR image not containing the TC center, only the coarsely estimated TC center position is obtained, and $D_1 = 82.9$ km. For Figs. 7(r), 8(d), and 12, the WIOTs are too weak in some regions where the retrieved SSWDs have low accuracy. For Fig. 7(y), the land-induced disturbance deformed the typical spiral pattern of the TC wind field. This causes the algorithm performance's degradation.

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